

Joshua Dinn

ANALYST

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EDUCATION

Monash University, Melbourne, Australia Mar 2023 – Oct 2027
Bachelor of Engineering (Software Engineering) and Bachelor of Commerce (Econometrics)
WAM: 79.97 (Distinction) | GPA: 3.48 / 4.0 | 100% in FIT2107 (Software Quality & Testing)

EXPERIENCE

Collingwood Football Club, Data & Analytics Intern Mar 2026 – Jun 2026

- Built a membership churn prediction model on historical transaction and engagement data to flag at-risk members, giving the commercial team a ranked list to work from rather than a hunch.
- Created an interactive dashboard breaking down churn by segment, using SHAP analysis to surface the behaviours driving retention so stakeholders could see the reasoning behind each flag.
- Set up a synchronisation workflow between the club's web and mobile app repositories to keep deployments consistent across teams.

Monash eSolutions, AI Fitness Trainer Jul 2026 – Present

- Run in-person sessions teaching students and staff how to use AI effectively, ethically, and in everyday study and work.
- Helped deliver a university webinar on AI to an audience of 200+ attendees.

Monash DeepNeuron, Optimised Computing Member Sep 2025 – Present

- Software team member on HarvestAI, an autonomous strawberry-picking robotic arm, working on the computer vision that detects fruit and classifies ripeness in real time.
- Teaching Assistant for an 8-week C++ and accelerated computing workshop run with NVIDIA, the Pawsey Supercomputing Centre, and NCI. NVIDIA CUDA certified.

Outlier AI, AI Trainer (Mathematics) Jun 2024 – Dec 2025

- Reviewed and corrected AI-generated mathematical solutions written in LaTeX to improve accuracy and reasoning quality.
- Wrote detailed feedback on computational and logical errors across advanced problem sets.

TMJ Tutoring, Mathematics Tutor Feb 2023 – Present

- Tutor high school students in mathematics, focusing on exam preparation and breaking hard problems into steps they can follow.

PROJECTS

Algorithmic Trading System for S&P 500 ETF May 2025

- Built and backtested a trading system using Hidden Markov Models to classify market regimes for SPY.
- Used RSI, ATR, and volatility signals to drive position sizing and trade execution.
- Deployed it on an AWS Linux server with the Alpaca API for automated paper trading.

Volatility Forecasting with a GARCH Model Apr 2025

- Built a volatility forecasting pipeline for the S&P 500 using a GARCH(1,1) model fitted by maximum likelihood on a decade of daily log returns.
- Validated the fit with Ljung-Box and ARCH-LM residual diagnostics, then forecast conditional variance 100 days ahead, automating data processing, evaluation, and reporting in Python.

OLS vs MLE: Monte Carlo Study of Estimator Performance Ongoing

- Designing a Monte Carlo study (Python) comparing OLS against correctly specified maximum likelihood estimators when the classical linear model assumptions fail: heteroscedasticity, skewed and fat-tailed errors, and binary or count outcomes.
- Measuring bias, relative MSE, and empirical coverage of nominal 95% confidence intervals across sample sizes from 25 to 2,000, with thousands of replications per design, to show where standard inference stops being trustworthy.

SKILLS & INTERESTS

Programming: Python, SQL, R, C++, MATLAB, JavaScript
Data & Quantitative: pandas, NumPy, XGBoost, statsmodels, Snowflake, Excel, OxMetrics
Analytical Methods: Regression and maximum likelihood estimation, time-series modelling, Monte Carlo simulation, forecasting, dashboarding and stakeholder reporting
Interests: Statistical modelling, financial econometrics, machine learning, data analysis